

Chemistry is the scientific study of the properties and behavior of matter.[1][2] It is a physical science within the natural sciences that studies the chemical elements that make up matter and compounds made of atoms, molecules and ions: their composition, structure, properties, behavior and the changes they undergo during reactions with other substances.[3][4][5][6] Chemistry also addresses the nature of chemical bonds in chemical compounds. In the scope of its subject, chemistry occupies an intermediate position between physics and biology.[7] It is sometimes called the central science because it provides a foundation for understanding both basic and applied scientific disciplines at a fundamental level.[8] For example, chemistry explains aspects of plant growth (botany), the formation of igneous rocks (geology), how atmospheric ozone is formed and how environmental pollutants are degraded (ecology), the properties of the soil on the Moon (cosmochemistry), how medications work (pharmacology), and how to collect DNA evidence at a crime scene (forensics).[9] Chemistry has existed under various names since ancient times.[10] It has evolved, and now chemistry encompasses various areas of specialisation, or subdisciplines, that continue to increase in number and interrelate to create further interdisciplinary fields of study. The applications of various fields of chemistry are used frequently for economic purposes in the chemical industry.[11]

Etymology Main article: Etymology of chemistry The word chemistry comes from a modification during the Renaissance of the word alchemy, which referred to an earlier set of practices that encompassed elements of chemistry, metallurgy, philosophy, astrology, astronomy, mysticism, and medicine. Alchemy is often associated with the quest to turn lead or other base metals into gold, though alchemists were also interested in many of the questions of modern chemistry.[12][13][14] The modern word alchemy in turn is derived from the Arabic word al-kīmīā (الكيمياء). This may have Egyptian origins since al-kīmīā is derived from the Ancient Greek χημία, which is in turn derived from the word Kemet, which is the ancient name of Egypt in the Egyptian language.[15] Alternately, al-kīmīā may derive from χημεία 'cast together'.[16]

Modern principles Laboratory, Institute of Biochemistry, University of Cologne in Germany The current model of atomic structure is the quantum mechanical model.[17] Traditional chemistry starts with the study of elementary particles, atoms, molecules,[18] substances, metals, crystals and other aggregates of matter. Matter can be studied in solid, liquid, gas and plasma states, in isolation or in combination. The interactions, reactions and transformations that are studied in chemistry are usually the result of interactions between atoms, leading to rearrangements of the chemical bonds which hold atoms together. Such behaviors are studied in a chemistry laboratory. The chemistry laboratory stereotypically uses various forms of laboratory glassware. However glassware is not central to chemistry, and a great deal of experimental (as well as applied/industrial) chemistry is done without it. Solutions of substances in reagent bottles, including ammonium hydroxide and nitric acid, illuminated in different colors A chemical reaction is a transformation of some substances into one or more different substances.[19] The basis of such a chemical

transformation is the rearrangement of electrons in the chemical bonds between atoms. It can be symbolically depicted through a chemical equation, which usually involves atoms as subjects. The number of atoms on the left and the right in the equation for a chemical transformation is equal. (When the number of atoms on either side is unequal, the transformation is referred to as a nuclear reaction or radioactive decay.) The type of chemical reactions a substance may undergo and the energy changes that may accompany it are constrained by certain basic rules, known as chemical laws.